Transform Pairs - Periodic Signals

Periodic Time Domain Signal $x(t) \overset{FS;\,\Omega_o}{\longleftrightarrow} X[k]$	Fourier Transform $X(\Omega) = 2\pi \sum_{k=-\infty}^{\infty} X[k]\delta(\Omega - k\Omega_o)$
$x(t) = \cos(p\Omega_o t)$	$X(\Omega) = \pi\delta(\Omega + p\Omega_o) + \pi\delta(\Omega - p\Omega_o)$
$x(t) = \sin(p\Omega_o t)$	$X(\Omega) = \frac{\pi}{j}\delta(\Omega + p\Omega_o) - \frac{\pi}{j}\delta(\Omega - p\Omega_o)$
$x(t) = e^{jp\Omega_o t}$	$X(\Omega) = 2\pi\delta(\Omega - p\Omega_o)$
$x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$	$X(\Omega) = \frac{2\pi}{T_s} \sum_{k=-\infty}^{\infty} \delta(\Omega - k\frac{2\pi}{T_s})$

Periodic Time Domain Signal	Discrete-Time Fourier Transform
$x[n] = \cos(\omega_o n), \quad \lvert\omega_o\rvert < \pi$	$X(e^{j\omega}) = \pi\delta(\omega + \omega_o) + \pi\delta(\omega - \omega_o), \quad \lvert\omega\rvert < \pi$
$x[n] = \sin(\omega_o n), \quad \lvert\omega_o\rvert < \pi$	$X(e^{j\omega}) = \frac{\pi}{j}\delta(\omega + \omega_o) - \frac{\pi}{j}\delta(\omega - \omega_o), \quad \lvert\omega\rvert < \pi$
$x[n] = e^{j\omega_o n}, \quad \lvert\omega_o\rvert < \pi$	$X(e^{j\omega}) = 2\pi\delta(\omega - \omega_o), \quad \lvert\omega_o\rvert < \pi$